

Ellen Su

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Education

Ph.D. Candidate, Center for Data Science

New York University

- Advisor: Todd Gureckis 2024 - present
- Relevant coursework: *mathematical statistics, computational linguistics, embodied visual learning*

B.S., Computer Science

Princeton University

Certificates in Applied Math and Cognitive Science

2019 - 2023

- Advisor: Tom Griffiths
- Thesis: *Revealing the Priors of Deep Learning Models Through Iterated Learning*

Publications

Su, E., Ho, M.K., Gureckis, T. (2025). Integration of Language and Experience via the Instructed Bandit Task. *In submission*.

Su, E., Vellore, A., Chang, A., Mura, R., Nelson, B., Kassianik, P., & Karbasi, A. (2024). Extracting Memorized Training Data via Decomposition. *ArXiv, abs/2409.12367*

Su, E., Arevalo, J., Carpenter, A., & Singh, S. (2024). MOTIVE: A Drug-Target Interaction Graph For Inductive Link Prediction. *Neural Information Processing Systems. (spotlight)*

Arevalo, J., Su, E., Ewald, J.D., van Dijk, R., Carpenter, A., & Singh, S. (2024). Evaluating batch correction methods for image-based cell profiling. *Nature Communications, 15*.

Research Experience

Robust Intelligence

San Francisco, CA

Research Intern, PI: Amin Karbasi

Summer 2024

- Developed a decomposition jailbreak to extract training data from production LLMs
- Wrote and published a first author preprint by the end of internship (August 2024)

Broad Institute of MIT and Harvard

Cambridge, MA

ML Research Associate, PIs: Anne Carpenter, Shantanu Singh

2023 - 2024

- Used graph neural networks to advance drug target interaction (DTI) discovery
- Produced a co-first author spotlight publication in NeurIPS (June 2024) and second author publication in Nature Communications (March 2024)

Princeton University

Princeton, NJ

Senior Thesis, PI: Tom Griffiths

2022 - 2023

- Used convolutional neural networks (CNNs) as learning agents in an iterated learning chain to identify potential inductive biases in the models
- Observed how machine priors manifested in their sequential decision making behavior
- Presented results at the Program in Applied and Computational Mathematics Symposium

Junior Paper, PI: Ben Raphael

2021 - 2022

- Contributed to a machine learning algorithm that predicts copy number variation mutations from spatial transcriptomics data

Fellowships and Awards

2023	Outstanding Student Teaching Award, Princeton Computer Science
2022	Grace Hopper Celebration Grant, Princeton Computer Science
2021	Undergraduate Research Summer Fellowship, Princeton University
2020	International Internship Program Research Grant, Princeton University

Work Experience

Robust Intelligence	San Francisco, CA
<i>Machine Learning Engineer Intern</i>	<i>Summer 2024</i>
• Researched and proposed implementation plan for multilingual support in AI firewalls • Built a multilingual dataset of prompt injections; benchmarked and evaluated multilingual language models in terms of performance and latency	
J.P. Morgan Chase & Co	Chicago, IL
<i>AI & Data Science Analyst</i>	<i>Summer 2022</i>
• Created a decision-making tool (with Python, SQL, Alteryx, and Tableau) for the email marketing team to optimize their campaigns • Picked up manager's responsibilities during an unexpected transition of leadership and trained the incoming associate	

Teaching Experience

Princeton Tutoring	Princeton, NJ
<i>Tutor</i>	<i>2021 - 2024</i>
• Teaching math, physics, and computer science to middle and high school students	
Princeton Computer Science	Princeton, NJ
<i>COS495 Undergraduate Course Assistant</i>	<i>Spring 2023</i>
• Assisted course staff, graded homework assignments, and answered student questions on course structure, assignments, and deliverables for Web3 and Blockchain course	

Service

Princeton First Aid and Rescue Squad	Princeton, NJ
<i>Emergency Medical Technician</i>	<i>2021 - 2023</i>
• Served as a first responder; stabilized, medicated, and transported patients • Certified by National Registry of Emergency Medical Technicians (NREMT)	

Skills and Languages

Interests	Machine learning, cognitive science, artificial intelligence
Languages (program)	Python (Pytorch, Tensorflow), Java, Javascript, C, SQL, CSS
Languages	English (native), Mandarin (native), Spanish (advanced)
Interests	Reading, running, cooking, and crosswords